

Inhibitor Tolerance: the Built Store Against the Plan

What the pooled chemogenomic store holds, which statements of the 031 plan survive the build, and the five decisions the store adds before a first run

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Abstract

The 031 plan for inhibitor-tolerance modeling was measured on dev-tree records. The pooled store the model trains on, build 001 of experiment 033, holds 6,394,540 served measurements as 6,042,771 cells, a cell being a strain in one environment, over 5,863 perturbed genes, 6,229 environments and 5,463 compounds. Every number here is read from that store.

The plan's data claims hold. All served records are present, the 648,977 Hoepfner pairs measured in both ploidy arms are there, the 41 Vanacloig reliability indexes match the plan's to floating-point error with a median ceiling of 0.84, the 71 repeated Hoepfner screen pairs give the plan's median reliability of 0.611, every dosed compound has a fingerprint vector, and the cross-source correlations are reproduced. Only Hillenmeyer needs its sign flipped.

Five facts of the store are new. The store's row is not the plan's row: the plan's statistics describe 3.08 million (gene, compound) pairs, and the store holds twice as many cells because a cell keeps dose, duration and ploidy arm. The label table holds the last measurement of each of 278,581 repeated cells, an accident of insertion order. No Hoepfner cell is paired across ploidy inside one environment, because the heterozygous arm ran 20 generations and the homozygous arm 5, so the data that would fit the heterozygosity factor changes duration with ploidy, and the two arms correlate at 0.13 against a noise limit of 0.61. 80,332 Hillenmeyer cells have no complete compound vector. And after per-source standardizing, 2 to 3 percent of each source's cells carry 41 to 80 percent of its squared error. Section 8 turns each into a decision with a recommendation, none of which has been run.

Section status:  ready  author-reviewed, keep stable  not done, agents may edit freely

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1 The question x

The planning documents for inhibitor-tolerance modeling, `notes-tex/031-inhibitor-tolerance-data` and `notes-tex/031-unified-representation`, measured everything on the dev-tree loaders, one row per served record, and most statistics on the (gene, compound) pair. A model does not train on those rows. It trains on the store the query built, build 001 of experiment 033, whose row is a strain in one environment. Whether the plan's numbers describe that store is an empirical question, and every answer here is read from the store itself.

Every script named below lives in `experiments/033-env-chemgen-pooled/scripts/`. `flatten_cells.py` reads each entry of the processed store once and writes its genotype, environment and every folded measurement as one table row (slurm 2988). `store_against_plan.py` re-measures each plan claim on that table and reads the plan's own values from the 031 result files at commit `bf1884c46`. `post_query_figures.py` and `notes_tex_post_query_tables.py` draw the figures and render the tables.

Vocabulary. A **cell** is one entry of the store: a genotype, its ploidy, and one environment. An **environment** is the content address the graph adapter gives a condition: medium composition, temperature, the dosed compounds at their concentrations, aerobicity and duration. The **functional dose** of a perturbed gene is the copies present over the copies in the reference, zero for a deletion and one half for a heterozygous deletion in a diploid. The **label** of a cell is the one number the store's label table holds for it. A **measurement** is one served record, and a cell folds every measurement that shares its genotype and environment. The **orientation** of a source is the factor, plus or minus one, that makes a negative response a fitness defect. A **pair** is one (gene, compound) combination with the response averaged over every dose, screen and ploidy arm that measured it, which is the unit the plan's joinability statistics use. The **plan** is the two 031 documents together. The **reliability** of a served value is the Pearson correlation over genes between two screens of the same environment, and its square root is the **ceiling** on the correlation any model can reach with the noise-free response. The **store** is build 001 of the pooled chemogenomic dataset, `/db/experiments/033-env-chemgen-pooled-001-pooled-build`.

Measured. Every axis of the store per source (Section 3); the ploidy arms and what pairs them (Section 4); the environment vocabulary, the dose and the compound vectors (Section 5); which measurement the label table holds and how well repeated screens agree (Section 6); the pooled target after orienting and standardizing (Section 7).

Not measured. No model has been trained on the store. Every statement about what a loss, a token or an operator would do with these numbers is labeled as a hypothesis, and Section 8 lists the decisions the measurements leave open.

2 The plan against the store x

Table 1 puts fifteen statements of the plan beside the store's value for the same quantity. The reading is computed, not judged: a row holds when the store's value is within the row's tolerance of the plan's, it differs when it is not, and it is new when the plan did not measure the quantity.

Table 1: The plan's statements against the built store. Counts are exact; the tolerance is zero for counts, 2 percent for the ceiling and the reliability, 10 percent for the spread ratio and 15 percent for the Vanacloig share. The plan's row count is the sum of its four (gene, compound) pair counts.

plan claim	plan	store	reading
records in the four kept datasets	6,394,540	6,394,540	holds
Vanacloig share of the pool, percent	2.2	2.4	holds
training rows, the plan's (gene, compound) pairs against store cells	3,084,000	6,042,771	differs
Hoepfner cells in both ploidy arms, same compound	648,977	648,977	holds
Hoepfner cells in both ploidy arms, same environment	not measured	0	new
widest over narrowest response sd	19.6	17.0	differs
sources whose sick tail is positive	HET	HET	holds
Vanacloig median ceiling over 41 compounds	0.84	0.84	holds
Vanacloig compounds with no signal	3	3	holds
Hoepfner repeated screen pairs	71	71	holds
Hoepfner median reliability of one served value	0.611	0.611	holds
label of a repeated cell	a read-time policy	the last measurement, 278,581 cells	new
dosed compounds with an FCFP4 vector	all	5,463 of 5,463	holds
cells with no complete compound vector	not measured	80,332	new
sources where a physical or medium field varies	none	HET, Hoepfner	differs

What holds. Nine rows hold, and the ones that matter most hold exactly. The store carries all 6,394,540 served records. The 648,977 Hoepfner pairs measured in both ploidy arms are there. All 41 Vanacloig reliability indexes agree with the plan’s to within floating-point error, so the median ceiling of 0.84 and the three compounds without signal are properties of the store and not only of the dev-tree records. The 71 repeated Hoepfner screen pairs and their median reliability of 0.611 are recovered from the folded measurements alone, without the raw score matrices the plan read them from. Every one of the 5,463 dosed compounds has a fingerprint vector. The dev-tree loaders and the served graph therefore describe the same data, which the plan assumed and had not tested.

What differs. Three rows differ and all three come from one cause, the unit of a row. The plan’s statistics describe 3,084,000 (gene, compound) pairs. The store holds 6,042,771 cells, because a cell keeps the dose, the duration and the ploidy arm that a pair averages away (Section 3). The spread ratio falls from 19.6 to 17.0 for the same reason. And two sources vary a physical or medium field inside themselves, which the plan noted for Hillenmeyer’s generation arms and did not carry into the representation (Section 5).

What is new. Three rows are quantities the plan did not measure, and each one blocks or changes a step of the implementation order. No Hoepfner cell is paired across ploidy inside one environment (Section 4). The label table holds the last measurement of each of the 278,581 repeated cells (Section 6). And 80,332 cells have no complete compound vector, so the processor step that raises on a missing vector would raise on them (Section 5).

3 The unit of a training row ✗

Table 2 counts what the store holds. The pool is 6,042,771 cells over 5,863 perturbed genes, 6,229 environments and 5,463 compounds. The gene count corrects an earlier note: 6,607 is the size of the S288C gene set the query filtered on and the width of the model’s gene axis, and 5,863 of those genes are perturbed in at least one cell. Vanacloig’s three host genes are among them, because the other three sources screen *PDR1*, *PDR3* and *SNQ2* as ordinary query genes.

Table 2: The store per source. Measurements are served records, cells are store entries, and the last three columns count cells whose environment doses no compound, doses two, or doses a compound that carries no InChIKey. The bottom row counts genes, environments and compounds once across the pool.

dataset	records	cells	genes	environ- ments	compounds	no com- pound	two compounds	no InChIKey
Vanacloig 2022	143,218	143,218	3,598	41	41	0	0	0
Hillenmeyer HET	2,698,797	2,591,199	5,825	455	290	57,388	147,953	22,944
Hoepfner 2014	3,124,319	2,880,165	5,842	563	148	0	0	0
Wildenhain 2015	428,206	428,189	242	5,170	5,170	0	0	0
All four	6,394,540	6,042,771	5,863	6,229	5,463	57,388	147,953	22,944

A cell is not a pair. The plan’s joinability table, its spread and skew per source, and its cross-dataset correlations are all computed on the pair: one gene under one compound, averaged over every dose, screen and ploidy arm. Table 3 computes the same statistics at both units from the store. Computed at the pair, the store reproduces the plan: the spread agrees to the third decimal for all four sources, and the pair counts differ by 596 for Hillenmeyer and 443 for Hoepfner out of 1.65 and 0.86 million. Computed at the cell, the two large sources change. Hillenmeyer has 1.57 cells per pair and Hoepfner 3.34, and their spread is 15 and 25 percent wider at the cell than at the pair, because averaging over doses and arms removes variance that a model trained on cells has to fit.

Table 3: The same response statistics at the store’s unit and at the plan’s. Pairs are (gene, compound) means computed from the store’s measurements; plan pairs and the plan columns are the 031 values. Vanacloig and Wildenhain have one cell per pair, so their columns coincide.

dataset	store cells	pairs	plan pairs	cells per pair	sd, cell	sd, pair	sd, plan	skew, cell	skew, plan
Vanacloig 2022	143,218	143,218	143,218	1.00	0.70	0.70	0.70	-2.57	-2.57
Hillenmeyer HET	2,591,199	1,651,844	1,651,248	1.57	0.44	0.38	0.38	1.56	1.24
Hoepfner 2014	2,880,165	861,788	861,345	3.34	1.62	1.29	1.29	-4.58	-4.44
Wildenhain 2015	428,189	428,189	428,189	1.00	7.50	7.50	7.50	-4.61	-4.61

The environment vocabulary. Table 4 gives the vocabulary an environment encoder sees. Three sources are dense: an environment carries about 3,500 cells in Vanacloig and about 5,700 in Hillenmeyer and Hoepfner, so an environment is a near-complete profile over the source’s genes. Wildenhain is sparse. Its 5,170 environments carry a median of 68 cells each, between 7 and 185, out of 242 genes, so a Wildenhain compound is a partial profile and no two compounds need share their genes. Hoepfner is the only source that revisits its compounds: a compound appears in a median of three environments and at most thirteen, and only one of its 148 compounds appears in a single environment.

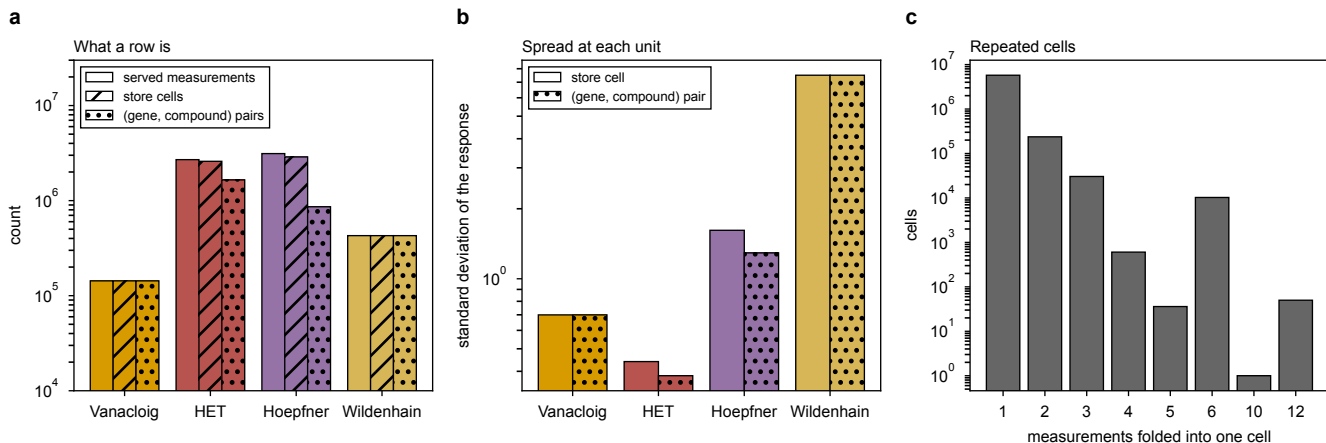


Figure 1: The store’s row is a cell, and two sources hold more cells than pairs. a, Served measurements, store cells and (gene, compound) pairs per source on a log scale; the gap between the second and third bar is what averaging over dose, duration and ploidy arm removes. **b**, Standard deviation of the response at the cell and at the pair. **c**, How many measurements a cell folds; 5,764,190 cells hold one and 278,581 hold more.

Table 4: Environments per source. Cells per environment is the minimum, median and maximum. Doses and environments per compound are the median and the maximum over the compounds dosed alone with an InChIKey.

dataset	environ- ments	cells per environment	compounds	doses per compound	environments per compound	one environment only
Vanacloig 2022	41	3,388 / 3,493 / 3,514	41	1 / 1	1 / 1	41
Hillenmeyer HET	455	5,677 / 5,677 / 5,802	290	1 / 7	1 / 10	221
Hoepfner 2014	563	4,402 / 5,697 / 5,773	148	2 / 7	3 / 13	1
Wildenhain 2015	5,170	7 / 68 / 185	5,170	1 / 1	1 / 1	5,170

4 The genotype axis and the ploidy arms $\times$

The functional dose takes the two values the plan expected and no others (Table 5). 1,826,549 cells sit at zero, a full deletion, and 4,216,222 at one half, a heterozygous deletion in a diploid. Seventy percent of the pool is therefore the perturbation kind the tensor layer cannot yet distinguish from a deletion.

Table 5: Genotype and dose per source. The functional dose is copies present over copies in the reference. The dose basis is counted over cells that dose one compound. The molar share is the cells whose concentration converts to molar without a molecular weight; the plan’s share is over records with a dosed compound, which is why Hillenmeyer’s is higher.

dataset	ploidy	cells at dose 0	cells at dose 1/2	dose basis	molar dose, percent of cells	plan, percent of records
Vanacloig 2022	haploid	143,218	0	IC30, fixed	0.0	0.0
Hillenmeyer HET	diploid	0	2,591,199	unstated, fixed	87.9	95.6
Hoepfner 2014	diploid	1,255,142	1,625,023	IC30	100.0	100.0
Wildenhain 2015	haploid	428,189	0	unstated	100.0	100.0

The two arms never share an environment. Hoepfner is the only source with both arms, and the plan rests the learned scale factor γ on its 648,977 pairs. The store confirms the count and adds a condition the plan did not state (Table 6). Matched on the compound alone, 648,977 pairs are measured in both arms. Matched on compound and concentration, 1,140,517 cells are. Matched on the environment, none is. Every heterozygous cell ran for 20 generations with no duration in hours stated, and every homozygous cell ran for 5 generations over 16 hours (Table 7). Duration is part of an environment’s identity, so the two arms are disjoint sets of environments, and in this source ploidy and duration are one variable.

The arms agree weakly. On the 1,140,517 cells matched on compound and concentration, the two arms correlate at Pearson 0.127 and Spearman 0.101. The repeat reliabilities of the arms, 0.576 and 0.639, put the limit for two measurements of the same quantity at 0.607, so the arms share 0.21 of what noise alone would allow. Hits overlap more than the correlation suggests: 8,000 of the 57,026 homozygous hits are also heterozygous hits, against 2,851 expected by chance. Figure 2c shows the shape behind both numbers. The mass lies along the two axes, cells that respond in one arm and sit at zero in the other, with a thin lower-left quadrant where both respond.

Table 6: Hoepfner cells measured in both ploidy arms, at three levels of matching. The limit is the correlation two noisy measurements of one quantity would show, the square root of the product of the two arms’ repeat reliabilities (Table 11). Per compound is the median Spearman over compounds with at least 100 paired genes. A hit is a paired cell in the bottom 5 percent of the homozygous arm, a shared hit is one that is also in the bottom 5 percent of the heterozygous arm, and the last column is the shared count expected if the two arms were independent.

matched on	cells in both	Spearman	Pearson	limit	Pearson over limit	per compound	hits	shared hits	shared by chance
same environment	0								
same compound and concentration	1,140,517	0.101	0.127	0.607	0.210	0.098	57,026	8,000	2,851
same compound	648,977	0.124	0.150	0.607	0.247	0.117	32,449	5,218	1,622

What this does and does not say about γ . The plan’s hypothesis is that a heterozygote attenuates the deletion response, which a scalar γ multiplying the operator can express. The measurement bounds that reading and cannot test it. A pure attenuation would make the two arms proportional and would put their correlation at the limit, and it sits at a fifth of the limit. Three things other than ploidy could produce the gap, and the store cannot separate them. The arms differ in duration, 20 generations against 5. Each screen is normalized to its own pool, so a score is relative to the other strains of its arm. And the heterozygous pool contains the essential genes the homozygous pool cannot, which changes what a median strain is. *Hypothesis (untested):* most of the gap is biology, because a heterozygous deletion exposes the drug’s target by dosage while a homozygous deletion exposes the pathways that buffer it. If so, γ as a scalar is the wrong form and the arm needs its own interaction with the compound.

5 The environment: fields, dose and compound vectors ✗

Physical and medium fields. The plan gives the physical fields no channel, on the ground that each is constant inside a source and the dataset token can carry it. Table 7 reads the fields from the store. The ground holds for Vanacloig and Wildenhain, where every field takes one value. It does not hold for the other two. Hillenmeyer varies three fields: 28,828 cells are in SD or YP rather than YPD, 28,560 state a temperature of 20, 23 or 37 C, and 183,702, which is 7.1 percent of the source, ran for a generation count other than 20. The first two are confined to the ten environments that dose no compound (Table 9), where the medium or the temperature is itself the condition; every Hillenmeyer cell that doses a compound is in YPD with no temperature stated. The generation count varies among the dosed cells as well. Hoepfner varies duration, and the variation is the ploidy arm (Section 4). A token per source assigns one vector to all of these.

Table 7: Physical and medium fields of the store’s cells, the distinct values per source ordered by cell count. The cell counts behind each value are in `results/store_axes.csv`.

dataset	base medium	temperature, C	aerobicity	duration, h	duration, generations	generations
Vanacloig 2022	SYNH3_MINUS	30.0	anaerobic	48.0	6.5	
Hillenmeyer HET	YPD, SD, YP	unstated, 23.0, 37.0, 20.0	aerobic	unstated	20.0, 5.0, 15.0, 0.0, 10.0	
Hoepfner 2014	YPD	30.0	aerobic	unstated, 16.0	20.0, 5.0	
Wildenhain 2015	SC	30.0	aerobic	18.0	unstated	

Dose. The store confirms the three dose regimes (Table 5). 5,585,863 cells, 92 percent of the pool, state a concentration that converts to molar. Vanacloig states none. The basis token is close to a second dataset token: every Hoepfner cell is on a target-inhibition basis, every Wildenhain cell states no basis, and Hillenmeyer states none on 2,380,154 of its single-compound cells and a fixed basis on 5,704. Only Vanacloig splits, with 136,237 cells dosed to a target inhibition and 6,981 at a fixed concentration, which are its two compounds benomyl and methyl methanesulfonate.

Cells a compound vector cannot describe. The plan’s processor resolves each dosed compound against the embedding table and raises when a vector is missing. Table 8 counts what that does on the store. Eleven encoders embed all 5,463 compounds, and under each of them 80,332 cells still have no complete vector. All are Hillenmeyer cells, in fourteen environments (Table 9): 57,388 cells in ten environments dose no compound, and 22,944 in four dose tunicamycin, a homologue mixture that carries no InChIKey. The 147,953 cells that dose two compounds are covered, since both vectors exist and the representation pools a set. The three-dimensional encoder leaves 406,318 cells uncovered, 6.7 percent of the pool, because 21 compounds have no conformer.

Table 8: Compound vectors against the store, per encoder. Compounds are distinct InChIKeys in the pool. A cell is covered when every compound it doses has a vector; the four right-hand columns are the covered share of each source’s cells, in percent.

encoder	compounds	cells without a vector	Vanacloig	HET	Hoepfner	Wilden- hain
chemberta2_mlm	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
chemberta2_mtr	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
ecfp4_bit	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
ecfp4_count	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
fcfp4_count	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
maccs	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
mol2vec	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
mole_static	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
molformer_xl	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
rdkit_2d	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
roberta_zinc_480m	5,463 of 5,463	80,332	100.00	96.90	100.00	100.00
unimol_v1	5,442 of 5,463	406,318	100.00	93.82	91.47	99.91

Table 9: The fourteen environments no compound vector can describe, all in Hillenmeyer HET. Ten dose no compound and differ from the source’s standard condition in medium or in temperature; four dose tunicamycin.

kind	compound	concentration	base medium	temperature, C	generations	cells
no InChIKey	tunicamycin	150 nM	YPD	unstated	5	5,734
no InChIKey	tunicamycin	600 nM	YPD	unstated	5	5,734
no InChIKey	tunicamycin	150 nM	YPD	unstated	15	5,738
no InChIKey	tunicamycin	600 nM	YPD	unstated	15	5,738
no compound			SD	unstated	0	5,783
no compound			SD	unstated	5	5,802
no compound			SD	unstated	10	5,771
no compound			YP	unstated	5	5,734
no compound			YP	unstated	15	5,738
no compound			YPD	20	20	5,677
no compound			YPD	23	5	5,734
no compound			YPD	23	15	5,738
no compound			YPD	37	5	5,734
no compound			YPD	37	20	5,677

6 The label of a repeated cell, and what repeats say about noise ✗

The label table holds the last measurement. 278,581 cells fold more than one measurement, and the build left the choice among them to read time. The label table has already made it. The loop that fills the table in `torchcell/data/neo4j_cell.py` visits the measurements of a cell in order and overwrites the label with each one, so the value that remains is the last. Table 10 confirms it on every repeated cell of the store: the label equals the last measurement in all of them, the first in none, and the mean in one cell of 170,966 by coincidence. The order of measurements inside a cell is the order the query returned them, which carries no meaning.

Table 10: Which measurement the label table holds for a repeated cell. The three label columns are the share of repeated cells whose label equals that measurement. Range over sd is the median spread between a cell’s measurements in units of the source’s standard deviation. The last column is the Pearson correlation between the label and the mean of the cell’s measurements.

dataset	cells	re- peated	percent	label is last	label is first	label is mean	range over sd	r, label and mean
Vanacloig 2022	143,218	0	0.00					
Hillenmeyer HET	2,591,199	107,598	4.15	1.00	0.00	0.00	0.49	0.81
Hoepfner 2014	2,880,165	170,966	5.94	1.00	0.00	0.00	0.36	0.85
Wildenhain 2015	428,189	17	0.00	1.00	0.00	0.00	0.08	0.32

The choice is not small where it applies. The measurements of a repeated cell differ by a median of 0.49 standard deviations in Hillenmeyer and 0.36 in Hoepfner, and the label correlates with the mean of the cell’s measurements at 0.81 and 0.85. Repeated cells are 4.6 percent of the pool. They are spread over 585 environments and concentrated in 50 of them, the ones where two screens share at least 100 genes, which hold 265,023 of the 278,581 (`results/pool_totals.csv`).

Repeat agreement recovers the plan’s ceilings. Two screens of one environment are parallel measurements, so their correlation over genes estimates the reliability of one served value. The plan computed this for Hoepfner from the raw score matrices. Table 11 computes it from the store’s folded measurements and finds the same 71 screen pairs and the same median of 0.611 over both arms, with ceilings of 0.759 and 0.799 against the plan’s 0.758 and 0.797. For Hillenmeyer the store holds 19 repeated environments, with a median Pearson of 0.352 and a median Spearman of 0.248. The plan’s raw replicate agreement for the same source is a Spearman of 0.255 over 108 conditions, read from the raw arrays. The two estimates agree, and both sit far below the reliability of 0.84 that the served standard errors imply, which is the plan’s reason for not using the served errors for this source.

Table 11: Reliability from repeated screens inside the store. A screen pair is two screens of one environment and ploidy arm with at least 100 genes in common. The ceiling is the square root of the median Pearson. Methotrexate contributes the most pairs in every row, 30 of Hoepfner’s 71.

dataset	arm	environ- ments	screen pairs	Pearson median	quartiles	Spearman median	ceiling	largest contributor
Hillenmeyer HET	heterozygous	19	19	0.352	0.13 to 0.48	0.248	0.594	methotrexate (3)
Hoepfner 2014	homozygous	16	36	0.576	0.34 to 0.66	0.379	0.759	methotrexate (15)
Hoepfner 2014	heterozygous	15	35	0.639	0.21 to 0.72	0.337	0.799	methotrexate (15)

The evaluation folds are in the store as planned. The first-round evaluation holds out one Vanacloig compound at a time. Table 12 lists the 41 folds as the store holds them: between 3,388 and 3,514 cells each, every cell with a served standard error, and a reliability index identical to the plan’s. The fold table the evaluation needs can therefore be built from the store alone.

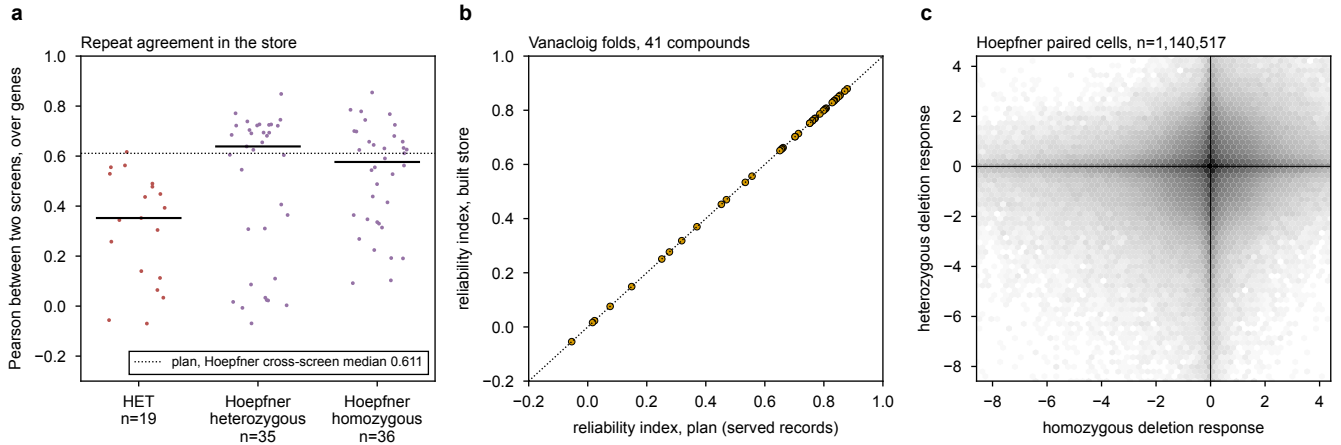


Figure 2: The store reproduces the plan’s ceilings, and the two ploidy arms barely agree. **a**, Pearson correlation over genes for every pair of screens that measured one environment, bar at the median; the dotted line is the plan’s Hoepfner median. **b**, Reliability index of each Vanacloig compound from the store’s standard errors against the plan’s value; all 41 lie on the diagonal. **c**, Hoepfner cells matched on compound and concentration, homozygous response against heterozygous response, density on a log scale; axes span the 0.5th to 99.5th percentile.

Table 12: The 41 compound-cold folds of Vanacloig, ordered by ceiling. Reliability is one minus the mean squared served standard error over the variance of the response across the fold’s genes; the ceiling is its square root, and zero where the index is negative.

compound	cells	basis	reliability	plan	difference	ceiling
methyl methanesulfonate	3,489	fixed	0.879	0.879	+0.000	0.938
2,2'-bipyridine	3,490	IC30	0.871	0.871	+0.000	0.933
crystal violet	3,388	IC30	0.855	0.855	+0.000	0.925
ferulic acid	3,500	IC30	0.850	0.850	+0.000	0.922
methylglyoxal	3,484	IC30	0.842	0.842	+0.000	0.918
acetovanillone	3,510	IC30	0.836	0.836	+0.000	0.914
2-propenamide, 3-(4-hydroxyphenyl)-, (2e)-	3,485	IC30	0.833	0.833	+0.000	0.913
ethanol	3,487	IC30	0.828	0.828	+0.000	0.910
1-ethyl-3-methylimidazolium chloride	3,492	IC30	0.808	0.808	+0.000	0.899
gamma-valerolactone	3,474	IC30	0.805	0.805	+0.000	0.897
acetosyringone	3,498	IC30	0.802	0.802	+0.000	0.895
ferulamide	3,490	IC30	0.798	0.798	+0.000	0.893
isobutanol	3,481	IC30	0.786	0.786	+0.000	0.887
furfural	3,494	IC30	0.771	0.771	+0.000	0.878
syringic acid	3,501	IC30	0.768	0.768	+0.000	0.877
2,6-dimethylpyrazine	3,505	IC30	0.762	0.762	-0.000	0.873
vanillic acid	3,486	IC30	0.761	0.761	+0.000	0.872
benomyl	3,492	fixed	0.753	0.753	+0.000	0.868
azelaic acid	3,497	IC30	0.752	0.752	+0.000	0.867
nonylacridine orange	3,507	IC30	0.714	0.714	-0.000	0.845
syringaldehyde	3,514	IC30	0.703	0.703	-0.000	0.838
4'-hydroxyacetophenone	3,507	IC30	0.702	0.702	+0.000	0.838
4-hydroxybenzaldehyde	3,493	IC30	0.662	0.662	+0.000	0.814
1-butyl-3-methylimidazolium chloride	3,485	IC30	0.658	0.658	+0.000	0.811
vanillin	3,506	IC30	0.656	0.656	+0.000	0.810
cinnamic acid	3,505	IC30	0.651	0.651	+0.000	0.807
myclobutanil	3,496	IC30	0.557	0.557	+0.000	0.746
2,4-dimethylimidazole	3,492	IC30	0.534	0.534	+0.000	0.731
benzoic acid	3,490	IC30	0.470	0.470	+0.000	0.685
caffeic acid	3,499	IC30	0.453	0.453	+0.000	0.673
sodium acetate	3,510	IC30	0.370	0.370	+0.000	0.608
sinapic acid	3,495	IC30	0.318	0.318	+0.000	0.564
acetamide	3,489	IC30	0.277	0.277	+0.000	0.526
4-hydroxybenzoic acid	3,510	IC30	0.251	0.251	+0.000	0.501
levulinic acid	3,491	IC30	0.149	0.149	+0.000	0.386
p-coumaric acid	3,510	IC30	0.076	0.076	-0.000	0.275
2-methylimidazole	3,489	IC30	0.023	0.023	+0.000	0.152
4-methylimidazole	3,501	IC30	0.016	0.016	+0.000	0.128
5-(hydroxymethyl)furfural	3,505	IC30	-0.054	-0.054	-0.000	0.000
sodium glyoxylate	3,491	IC30	-1.018	-1.018	+0.000	0.000
sodium butyrate	3,490	IC30	-0.467	-0.467	-0.000	0.000

7 The pooled target after the transform x

The plan orients each source so that negative is a defect and standardizes it by its own training-split statistics. Table 13 applies that transform to the store, with whole-source statistics standing in for the training split, and measures what it fixes and what it leaves.

Table 13: The pooled target per source. o is the orientation factor. Raw squared error is the source’s share of the pool’s squared deviation from each source’s own mean, before any standardizing. The z columns are quantiles of the oriented, standardized response. Robust sd is 1.4826 times the median absolute deviation.

dataset	o	cells, percent	raw squared error, percent	excess kurtosis	z at 0.1 percent	z at 99.9 percent	beyond 3 sd, percent	their squared error, percent	sd over robust sd
Vanacloig 2022	+1	2.4	0.2	19.8	-7.6	4.7	2.5	61.1	2.47
Hillenmeyer HET	-1	42.9	1.6	15.5	-7.6	5.0	1.7	40.5	1.60
Hoepfner 2014	+1	47.7	23.4	61.2	-10.3	4.7	1.9	63.8	5.21
Wildenhain 2015	+1	7.1	74.9	22.9	-7.0	1.1	3.1	80.0	6.69

What standardizing fixes. Without it the loss is Wildenhain’s. That source is 7.1 percent of the cells and 74.9 percent of the raw squared error, while Vanacloig, the application target, is 2.4 percent of the cells and 0.2 percent of the squared error (Figure 3b). The plan stated this qualitatively and the store gives it a size.

What standardizing leaves. Dividing by the standard deviation equalizes the second moment and nothing else, and the four distributions are far from a common shape (Figure 3a). The excess kurtosis runs from 15.5 to 61.2. The standard deviation is 1.6 to 6.7 times the robust standard deviation, so which scale is divided out changes the relative scale of the sources by up to a factor of four. And inside every source the squared error belongs to a small minority: between 1.7 and 3.1 percent of cells lie beyond three standard deviations, and they carry between 41 and 80 percent of their source’s squared error (Figure 3c). The sick tail reaches 7 to 10 standard deviations at the 0.1th percentile in all four sources, while the resistant tail reaches 5 in three of them and 1 in Wildenhain.

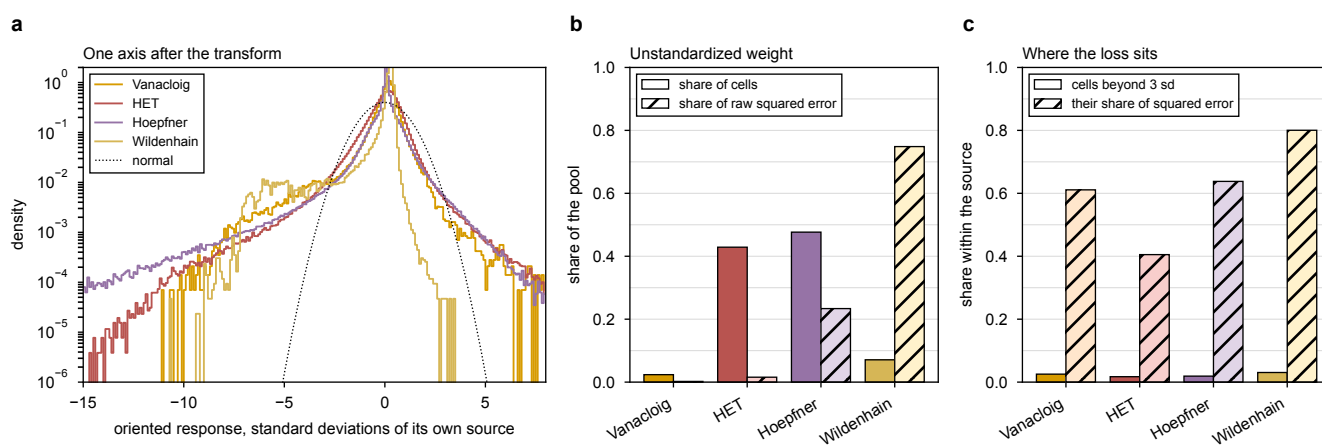


Figure 3: Standardizing removes the scale disagreement and leaves the shape disagreement. **a**, Density of the oriented response in standard deviations of its own source, log scale, with a standard normal for reference. **b**, Each source’s share of the pool’s cells beside its share of the raw squared error. **c**, The share of each source’s cells that lie beyond three standard deviations beside the share of that source’s squared error they carry.

Hypothesis (untested): under a squared-error loss on this target the model is fit mostly to the hits, and a per-compound correlation, which is how the evaluation scores, weights every gene of a fold equally. Whether the mismatch costs anything is a first-round question, and the measurement says where to look, not what will be found.

Cross-source agreement is unchanged. Table 14 repeats the plan’s agreement test on the store’s measurements for the six pairs of the four kept sources. Five pairs reproduce the plan’s count and correlation exactly, and the sixth, Hillenmeyer against Hoepfner, differs by 66 pairs in 170,903 and in the third decimal. The plan’s conclusion stands on the store: every pair correlates positively once oriented, and none above 0.12.

Table 14: Agreement between sources on the (gene, compound) pairs both measure, Spearman as served and after orienting, beside the plan’s values.

a	b	genes	compounds	pairs	plan pairs	pairs as served	oriented	plan, oriented
Vanacloig	HET	3,571	2	6,852	6,852	-0.023	+0.023	+0.023
Vanacloig	Hoepfner	3,592	2	6,977	6,977	+0.013	+0.013	+0.013
Vanacloig	Wildenhain	153	10	761	761	+0.078	+0.078	+0.078
HET	Hoepfner	5,805	30	170,903	170,837	-0.118	+0.118	+0.118
HET	Wildenhain	238	123	17,218	17,218	-0.018	+0.018	+0.018
Hoepfner	Wildenhain	237	34	4,787	4,787	+0.061	+0.061	+0.061

8 What the measurements change in the plan ✗

The plan’s data claims survive the build. Its implementation order meets five facts of the store that it did not anticipate, and each is a decision to take before the first run. The recommendations below are proposals. None has been run.

1. Choose the label of a repeated cell. The label table’s choice, the last measurement, follows from insertion order (Section 6). The alternatives are the mean of a cell’s measurements, which is the lower-noise target, or one measurement drawn per epoch, which keeps the noise and shows it to the model. *Recommendation:* the mean, with the measurement count kept as a column so a loss can weight by it. The change is in the label-table loop and in nothing else, and it touches 278,581 cells.

2. Decide what a vectorless cell is. 80,332 Hillenmeyer cells have no complete compound vector (Section 5), and the plan’s processor raises on them. Two routes exist. The cells can be left out of the first round, which costs 1.3 percent of the pool and removes the ten medium and temperature conditions, which are outside the compound question, together with tunicamycin. Or the environment context can carry a learned vector for the absence of a compound, which keeps them and adds a parameter whose meaning is mixed. *Recommendation:* leave them out of the first round by a read-time selection, and record the selection beside the run.

3. Give the token the arm, or give the model the duration. A token per source cannot carry a field that varies inside the source (Table 7). For Hoepfner the variation is exactly the ploidy arm, so a token per source and arm, five tokens in all, carries it at no cost. For Hillenmeyer the medium and the temperature stop varying once the vectorless cells of decision 2 are left out, and the generation count remains. It is a number, so a scalar channel for the generation count is the direct encoding. *Recommendation:* both, since the first is a vocabulary entry and the second is one masked scalar beside the dose.

4. Read the γ arm as ploidy and duration together. The only data that can fit γ changes duration with ploidy (Section 4), and the two arms correlate at a fifth of their noise limit. A learned scalar γ fit on these cells is an estimate of the joint effect, and a small or unstable γ would not show that heterozygosity is unimportant. *Recommendation:* keep the fixed-identity arm as planned, and add to the learned arm a variant in which the compound rows of the operator are conditioned on the arm, so the round can tell a scale from an interaction. Hillenmeyer’s heterozygous cells at 5 and 20 generations, 86,078 and 2,407,497 of them, are the one place duration varies with ploidy held fixed, and they can be used to estimate the duration effect separately.

5. Decide whether the loss should follow the tails. Standardizing by the standard deviation leaves 41 to 80 percent of each source’s squared error on 2 to 3 percent of its cells (Section 7). That may be what is wanted, since the hits are the phenotype. It is a choice either way, and it interacts with the scale: dividing by a robust scale instead raises the weight of the heavy-tailed sources by up to a factor of four in scale. *Recommendation:* run the first round as planned on the squared error, log the per-source loss split into cells within and beyond three standard deviations, and hold a Huber loss as the second-round arm if the tail share of the loss tracks the raw share.

What needs no change. The record population, the orientation factors, the per-source standardization, the 41-fold compound-cold evaluation with its ceilings, the choice of the feature-class fingerprint as the default encoder, and the expectation that pooling buys a gene prior and not more labels for the same task are all as the plan has them, now measured on the store.

What is still missing. The tensor layer reads only a gene name. The store carries the copy numbers, the ploidy, the compounds, the doses and the durations in every cell’s record, and the cell table written here holds them as columns, so the processor changes of the plan’s implementation order are read-side work and need no further build.